

भारतीय मानक
Indian Standard

IS 9873 (Part 9) : 2017

खिलौनों की सुरक्षा

भाग 9 खिलौनों और बच्चों के उत्पादों में कुछ थैलेट एस्टर

Safety of Toys

Part 9 Certain Phthalates Esters in Toys and
Children's Products

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भारतीय मानक ब्यूरो
BUREAU OF INDIAN STANDARDS

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Price Group 1

Plastics Sectional Committee, PCD 12

FOREWORD

This Indian Standard (Part 9) was adopted by the Bureau of Indian Standards, after the draft finalized by the Plastics Sectional Committee had been approved by the Petroleum, Coal and Related Products Division Council.

Phthalates are used as plasticizers (softening agents) in the manufacture of soft vinyl [also known as polyvinyl chloride (PVC)], which in turn is used in toys and childcare articles. Phthalates do not bind to the soft vinyl, but are present as mobile components of the vinyl. However, the mere presence of phthalates in soft vinyl does not in itself equate to a health risk. It is the amount of phthalates that leach out of soft vinyl and are absorbed into the body that can be harmful. Although exposure to phthalates in soft vinyl through dermal contact or inhalation is negligible, phthalates may leach out of soft vinyl during periods of sustained mouthing action (sucking and chewing, but not licking) and enter the body through the saliva. Once in the body, some phthalates have the potential to cause adverse effects on reproduction system and development of child.

Earlier, the safe limits and test method for certain phthalates in toys were prescribed in IS 9873 (Part 3) : 1999 through Amendment 3. The Committee has now decided to publish a separate Part under IS 9873 for the requirements of certain phthalates and to adopt ISO 8124-6 : 2014 under dual numbering system for test methods for determination of phthalates.

This standard, IS 9873 has various parts under general title 'Safety of toys'. Other parts in this series are:

- Part 1 Safety aspects related to mechanical and physical properties
- Part 2 Flammability
- Part 3 Migration of certain elements
- Part 6 Determination of certain phthalate esters in toys and children's products

Further, the Committee has decided to formulate following new parts of IS 9873 which are under preparation:

- Part 4 Swings, slides and similar activity toys for indoor and outdoor family domestic use
- Part 5 Determination of total concentration of certain elements in toys
- Part 7 Requirements and test methods for finger paints
- Part 8 Age determination guidelines

Indian Standard

SAFETY OF TOYS

PART 9 CERTAIN PHTHALATES ESTERS IN TOYS AND CHILDREN'S PRODUCTS

1 SCOPE

This standard specifies maximum acceptable levels of certain phthalate esters in toys and children's products, except material not accessible.

2 REFERENCE

The standard given below contains provision, which through reference in this text, constitutes provision of this standard. At the time of publication, the edition indicated was valid. All standards are subject to revisions, and parties to agreements based on this standard is encouraged to investigate the possibility of applying the most recent edition of the standard indicated below:

<i>IS No.</i>	<i>Title</i>
9873 (Part 6)	Safety of toys: Part 6 Determination of certain phthalate esters in toys and children's products (<i>Under Print</i>)

3 MAXIMUM ACCEPTABLE LEVELS OF PHTHALATES

When tested in accordance with IS 9873 (Part 6), the toy products shall not show the presence of phthalates beyond the limits given as under:

<i>Characteristic</i>	<i>Requirement</i>
Vinyl in toys or childcare article	≤ 0.1 percent (individual or in combination) of: a) Bis (2-ethyl hexyl) phthalate (DEHP) b) Dibutyl phthalate (DBP) or c) Benzyl butyl phthalate (BBP)
Vinyl in any part of the toy or childcare article that can be placed in mouth of a child under 4 years of age	≤ 0.1 percent (individual or in combination) of: a) Di-isononyl phthalate (DINP) b) Di-isodecyl phthalate (DIDP) or c) Di- <i>n</i> -octyl phthalate (DNOP)

4 BIS CERTIFICATION MARKING

The product may also be marked with the Standard Mark.

4.1 The use of the Standard Mark is governed by the provisions of the *Bureau of Indian Standard Act*, 1986 and the Rules and Regulations made thereunder. The details of conditions under which the licence for the use of the Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

5 MARKING

5.1 The following shall be marked legibly and indelibly on each toy:

Name of manufacturer and/or his authorized representative and/or registered trade-mark, if any.

5.2 The following shall be marked legibly and indelibly on toy packing:

- Name of the Article;
- Country of origin;
- Manufacturer's or his authorized representative's name and address and trade-mark, if any;
- Batch No./Code No.;
- Month and year of manufacture;
- Instructions, for use and storage as applicable (on packing/leaflet/tag, etc);
- Safety directions, if any;
- Other provisions of Regulatory Authority, as required; and
- Usage's Age Group.

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Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of 'BIS Catalogue' and 'Standards : Monthly Additions'.

This Indian Standard has been developed from Doc No.: PCD 12 (10457).

Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

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